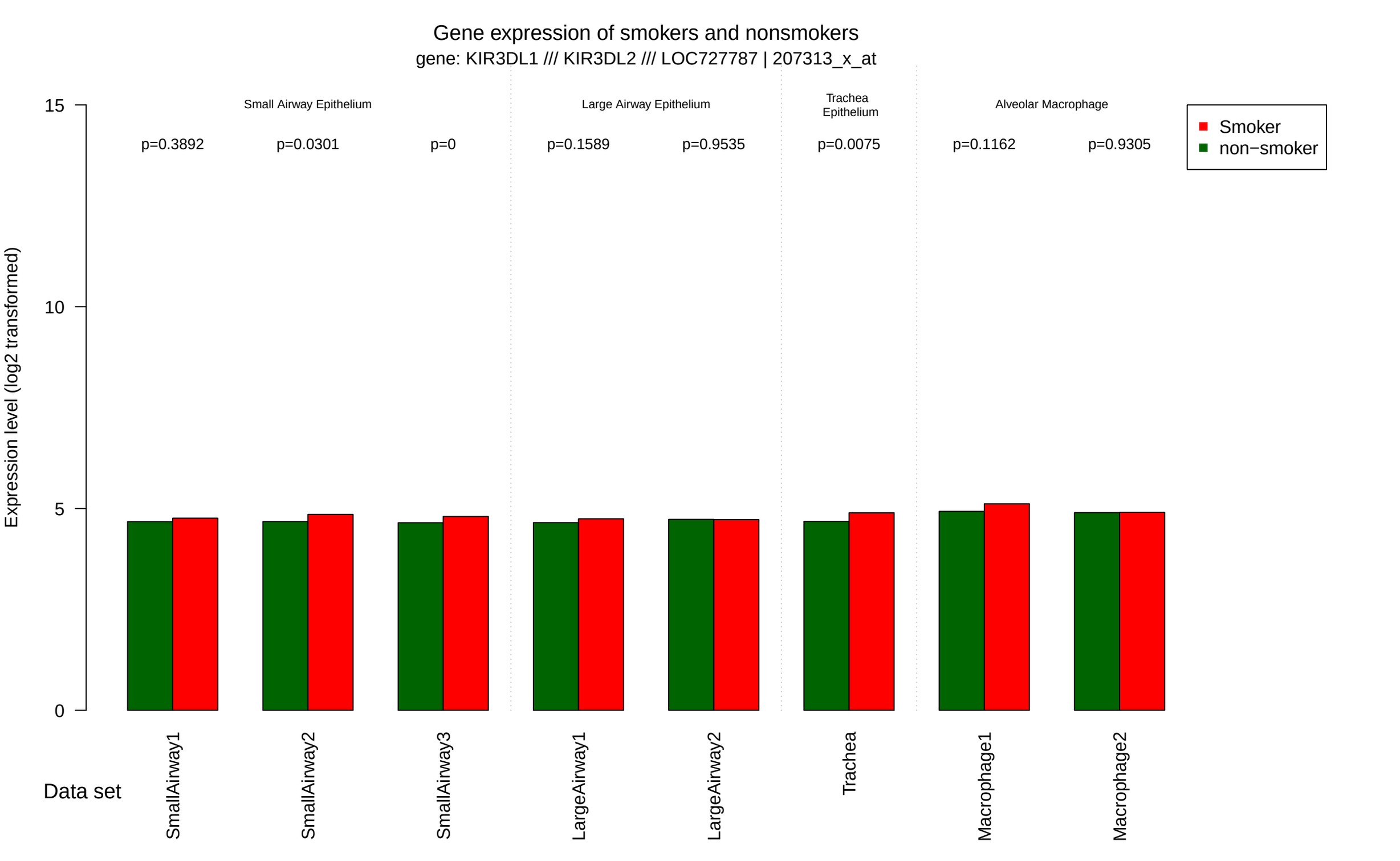
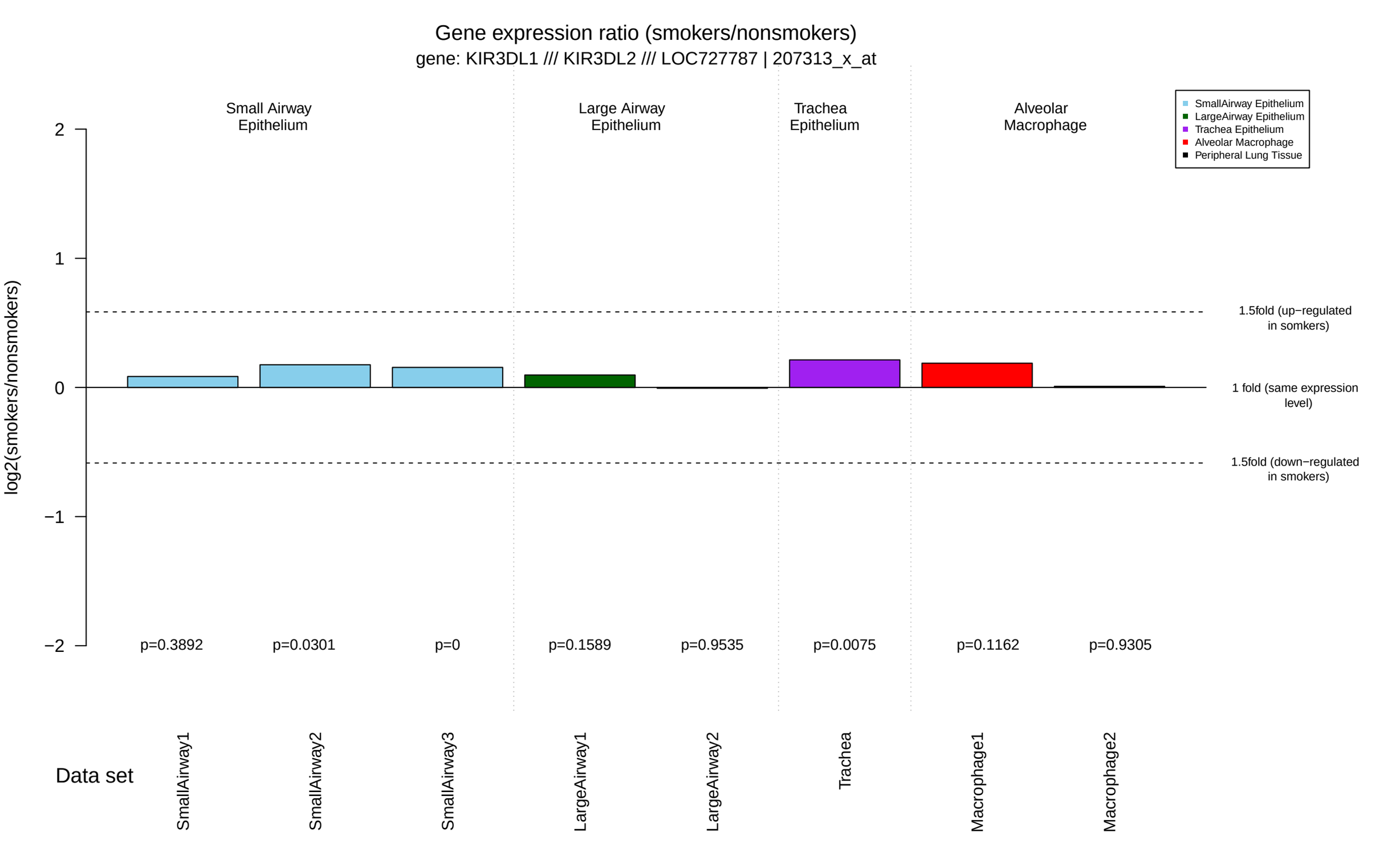
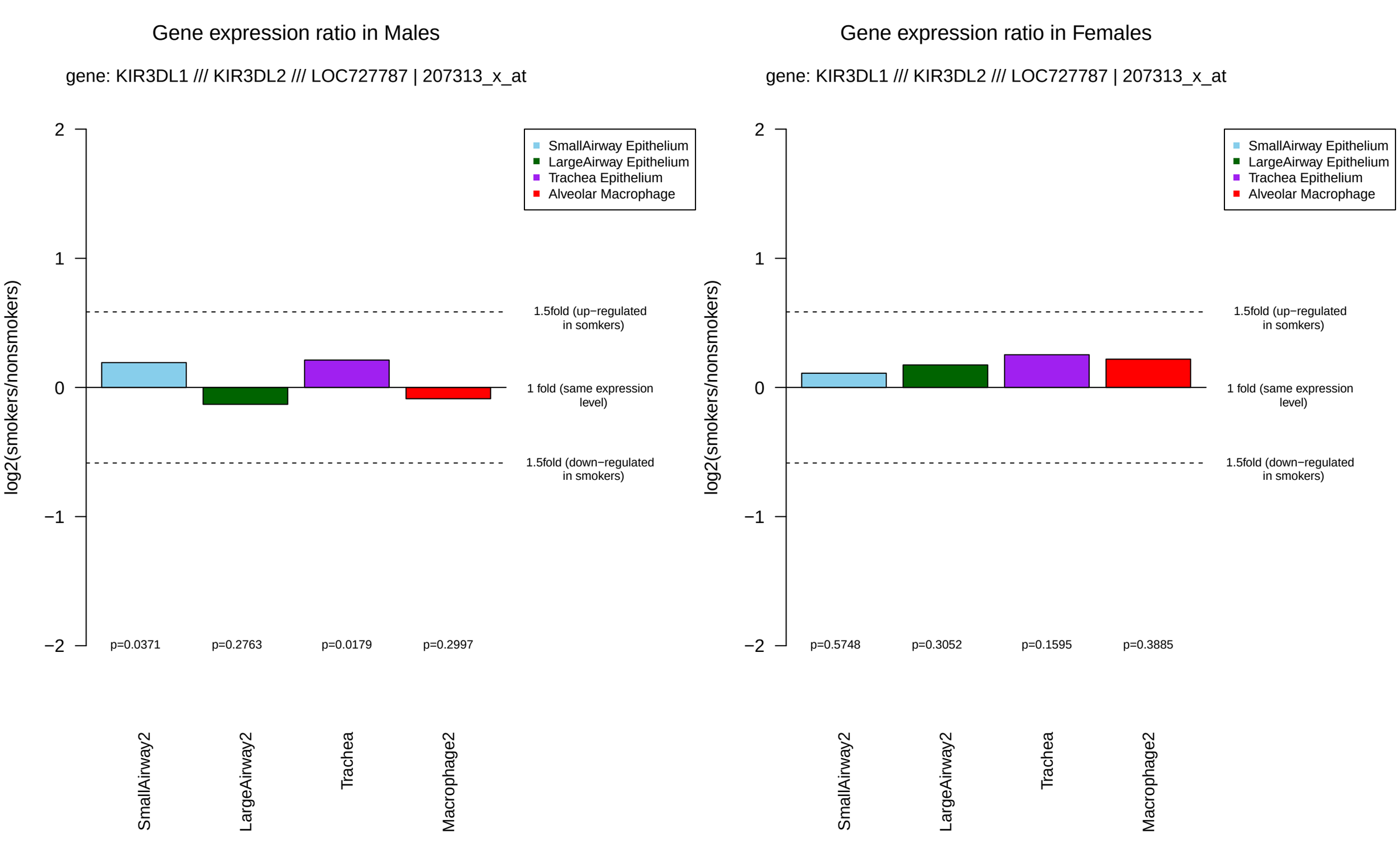
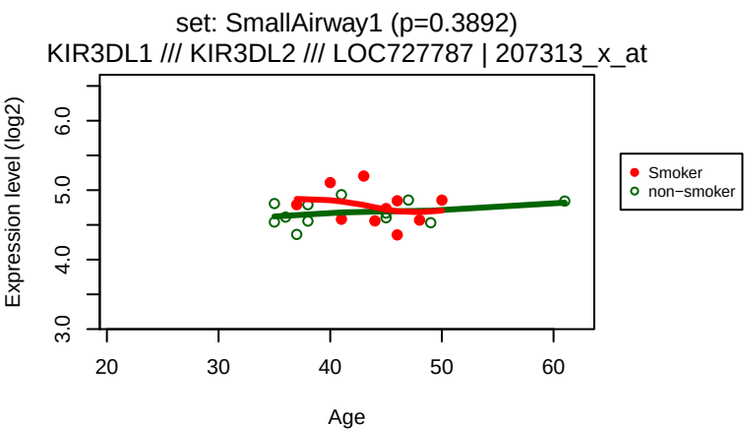
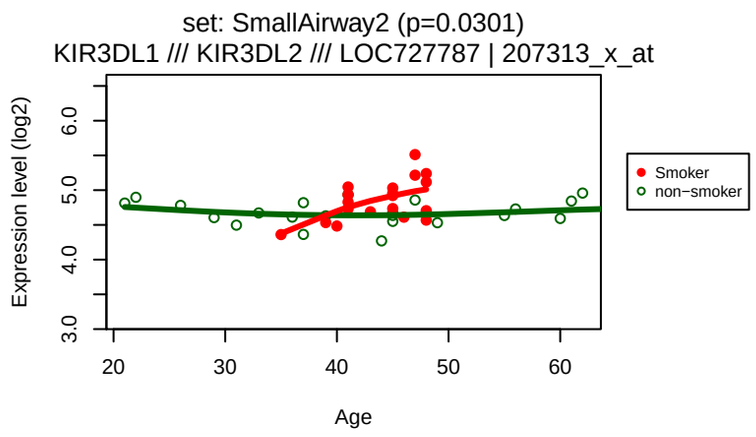
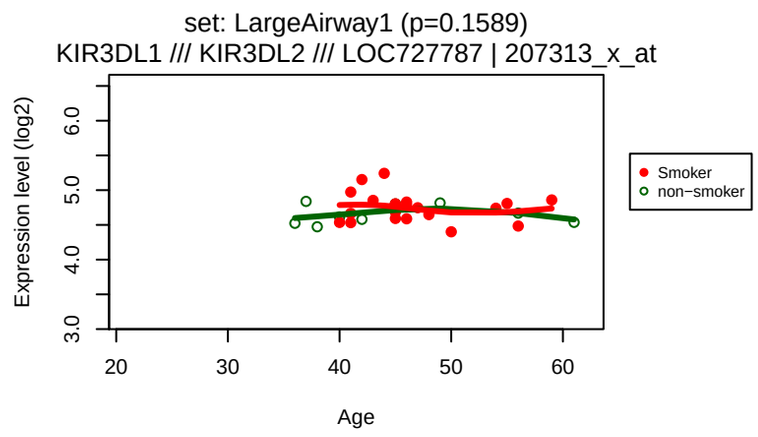

Gene expression of smokers and nonsmokers
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3892
p=0.0301
p=0
p=0.1589
p=0.9535
p=0.0075
p=0.1162
p=0.9305
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

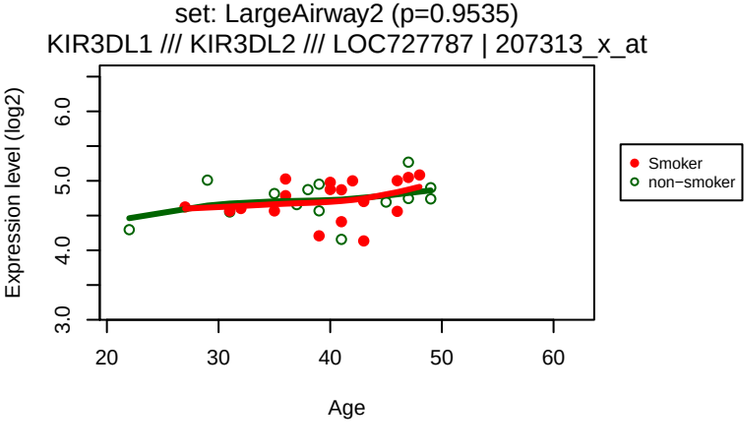

Gene expression ratio (smokers/nonsmokers)
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3892
p=0.0301
p=0
p=0.1589
p=0.9535
p=0.0075
p=0.1162
p=0.9305
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

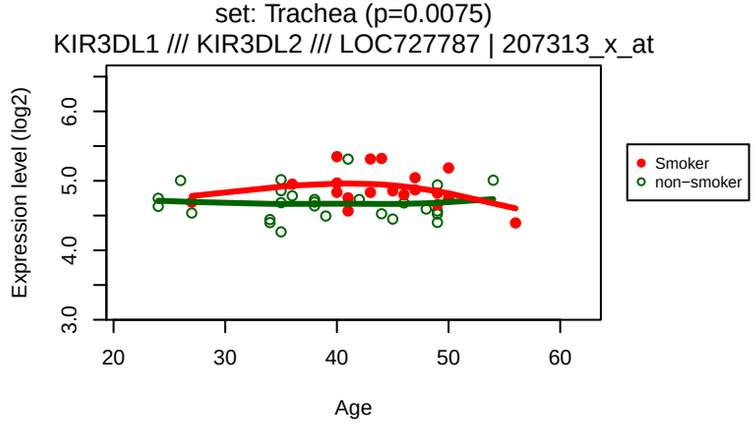

Gene expression ratio in Males
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0371
p=0.2763
p=0.0179
p=0.2997
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5748
p=0.3052
p=0.1595
p=0.3885
SmallAirway2
LargeAirway2
Trachea
Macrophage2

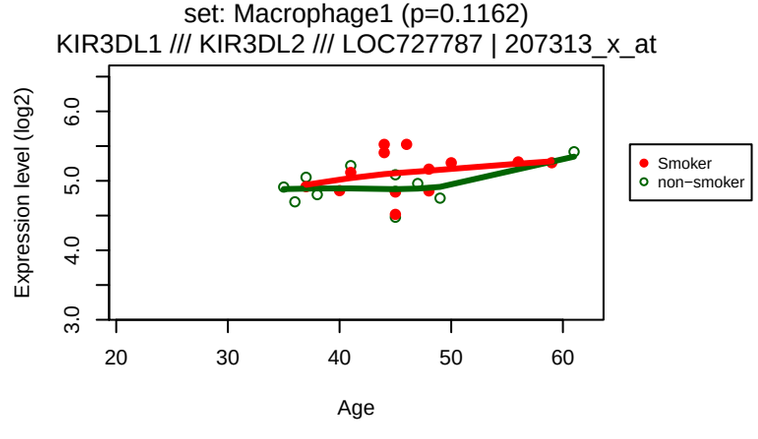

set: SmallAirway1 (p=0.3892)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker

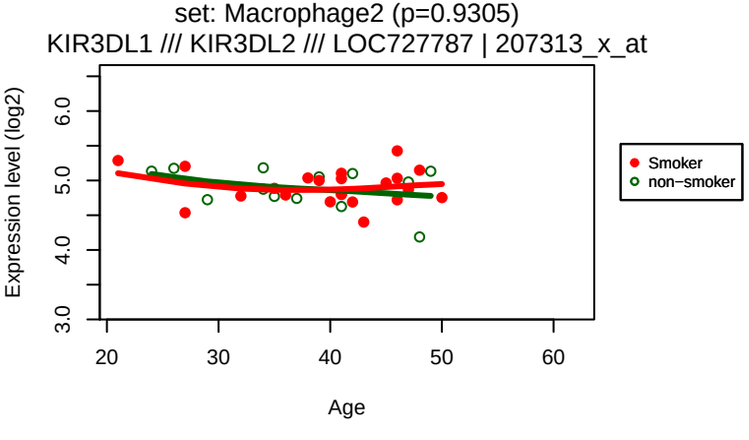

set: SmallAirway2 (p=0.0301)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker

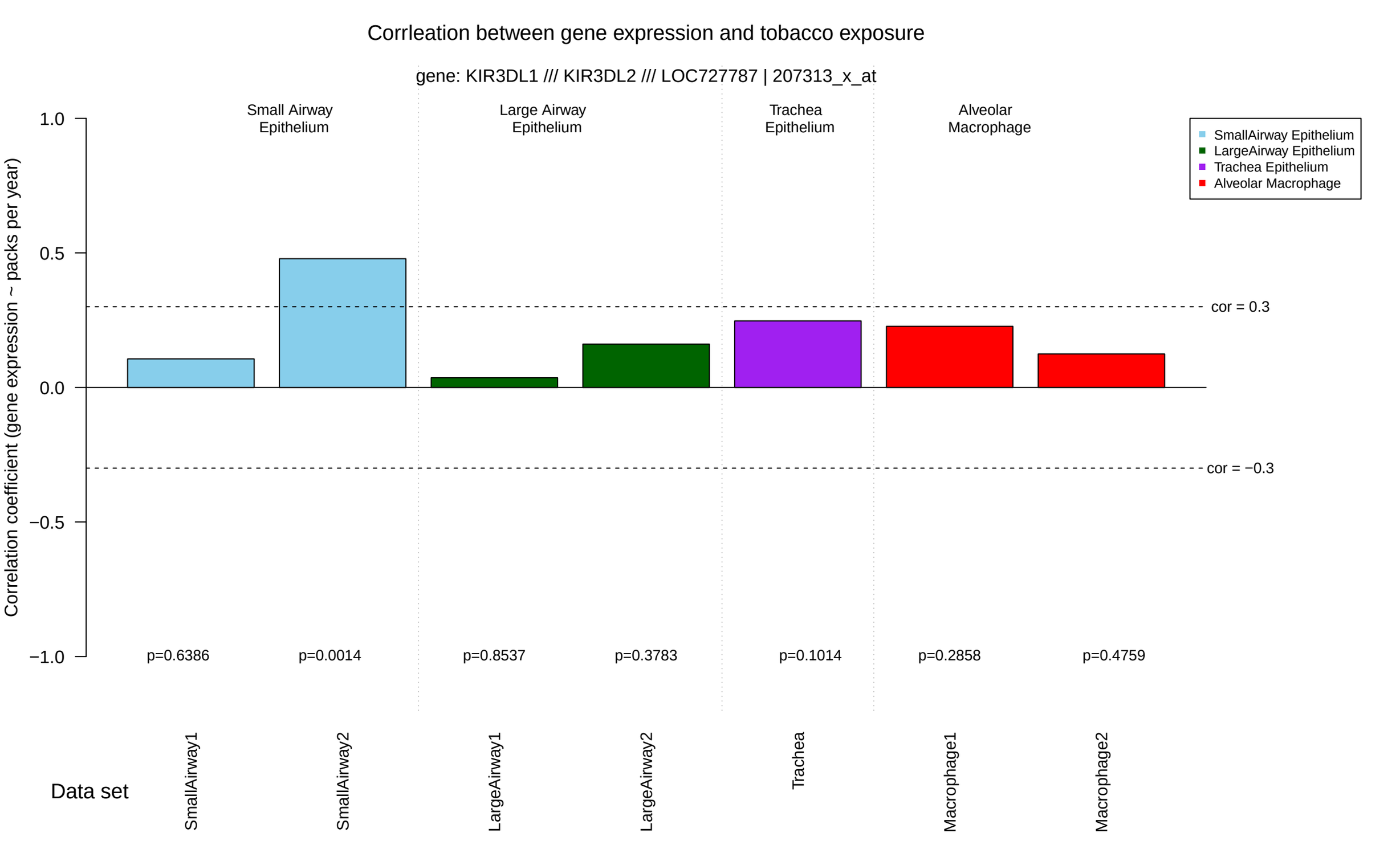

set: LargeAirway1 (p=0.1589)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9535)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0075)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1162)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9305)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 207313_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6386
p=0.0014
p=0.8537
p=0.3783
p=0.1014
p=0.2858
p=0.4759
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2